[CSE499A(10) - Group#5] Weekly Report: ShelfSense: AI-Based Retail Merchandising Audit

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III. Weekly Report 3

A. Follow-up from last week

Status of the targets set in Weekly Report 2:

1) SKU-110K sample and simulated videos — Replaced. We obtained three real phone videos of branded beverage fridges in Bangladeshi shops and used them instead of simulated video. The SKU-110K download moved to Sprint 4, where it is used to train the detector.
2) Kaggle notebook for Qwen2-VL-2B — Done. The notebook runs Qwen2-VL-2B-Instruct [1] zero-shot in fp16 on a free Kaggle T4 GPU (Section III-B3).
3) Prompts for the audit tasks — Done. Five prompts: fridge brand, product brands, counts per brand, empty space and price tags, each with a JSON answer format.
4) Evaluation set — Partly done. A ground-truth file lists the fridge brand, the product brands, empty space and price tags for each video. Per-brand counts and the brands in the third video are still unverified, so all numbers below use a draft ground truth.
5) Baseline results and failure cases — Done. Two prompt versions were run and scored automatically (Tables I and II).
6) Video VLMs in the literature table — [status].

B. Work narrative of the current week

1) Refined problem: verifiable audits: Our team's knowledge of Bangladeshi retail sharpened the problem. Brand companies (for example Coca-Cola, Pepsi/Transcom and Akij) give shops branded fridges on condition that only their products are stored inside, and pay extra for prime placement. Field auditors are sometimes bribed by shopkeepers and report false compliance. ShelfSense should therefore replace the auditor's word with evidence: every finding must point to a keyframe and timestamp a person can check. Two consequences shape the rest of the project:

- New key KPIs: cooler purity (only the owner's brands in a branded fridge) and paid placement (the agreed brand on the agreed shelf).
- Precision over recall for violations: a false "competitor in your fridge" accuses an honest shopkeeper, so invented brands are unacceptable.

2) Real footage and a new blur metric: Our three fridge videos are short WhatsApp clips (576×1024 portrait, 2.3 s, 6.4 s and 2.4 s). On real footage the adaptive Laplacian threshold from Report 2 failed: the Laplacian variance depends on scene content, so sharp close-ups scored 49 while motion-blurred frames scored 147, and no threshold separated them. We replaced it with the no-reference blur effect of Crete et al. [2], which compares a frame with a re-blurred copy of itself and gives 0 for sharp and 1 for fully blurred images. Real frames scored 0.24–0.39 and artificially blurred copies 0.56–0.72, so we drop frames above 0.5. The Laplacian score is still logged for comparison. At 4 fps the three clips gave 45 sampled frames, 30 kept and 15 duplicates; they contain no real motion blur.

3) Baseline VLM: The Kaggle notebook asks every question in two modes: image mode (each keyframe separately, answers combined per video) and video mode (up to 8 keyframes given to the model as one video). An evaluation script parses the JSON answers (tolerating code fences and extra text), normalises brand spellings and scores them against the ground truth. Cooler purity is computed from the answers: a violation is any product brand owned by a different company than the fridge brand. Version 2 of the prompts targeted each version 1 failure: look only behind the glass door, list each brand once, do not guess common brands, ignore shop signs, and give explicit empty answers. Model, image size and generation settings stayed fixed, so the versions differ only in the prompts.

TABLE I
Baseline Qwen2-VL-2B results on three fridge videos (image / video mode, draft ground truth).

Metric	Prompt v1	Prompt v2
Unparseable answers	22 / 2	8 / 0
Mean time per answer (s)	3.6	2.9
Fridge brand accuracy	0.67 / 0.33	0.67 / 0.67
Brand precision	0.27 / 0.30	0.13 / 0.25
Brand recall	1.00 / 0.50	1.00 / 1.00
Empty space accuracy	0.33 / 0.67	0.67 / 0.67
Price tag accuracy	0 / 0	0.33 / 0

TABLE II
Typical failure cases of the zero-shot VLM.

Failure	Example
Invented brands	Pepsi, 7up and Dr Pepper listed in a Coca-Cola fridge
No real counting	Every count 10 (or 1) in v1; 10/10/10 or 0/0/0 in v2
Invented price tags	Sequences such as 1, 2, 3 or TK 10, TK 15; the <price> placeholder copied
Repetition loops	The same brand repeated until the 256-token limit
Wrong text read	Shop sign or "property of" text given as the fridge brand; "nojo" for Mojo
Limits become targets	"At most 8 brands" → exactly 8 brands; "at most 5 prices" → 5 invented prices

4) Results: Table I shows that better prompts fixed the format problems: unparseable answers fell from 24 to 8, repetition loops stopped and answers became faster. They did not fix perception: brand precision got worse (0.27 → 0.13 in image mode) because the model still listed brands that are commonly sold rather than brands it could see, and counts were never real (Table II). With three videos these numbers show failure patterns, not significant differences.

5) Challenges faced:

- The blur filter from Report 2 did not transfer to real video. Solved by switching to the Crete blur effect, which is far less dependent on scene content.
- Too little real data. Three clips of 2–6 s cannot give reliable statistics; longer recordings (15–30 s per fridge) are needed.
- VLM hallucination. Prompting cannot stop the model from inventing brands, which is unacceptable for an audit that may accuse a shopkeeper.

6) Progress made: Sprint 3 is complete: the baseline VLM is running, scored automatically and its failure cases are documented. The results confirm the hybrid design of Report 1: a detector must do the counting, and the VLM should only be asked about one product crop at a time. We stopped prompt tuning.

C. Target for next weekly update

1) Record more and longer real footage.
2) Download SKU-110K and train a YOLO nano product detector; report mAP.
3) Select and annotate keyframes from our own footage for detector evaluation.
4) Test the VLM on single-product crops instead of whole frames.
5) Verify the draft ground truth (brands in the third video, per-brand counts).

D. Individual contribution

TABLE III
Contribution summary.

Task List	Member Assigned	Comments
Problem refinement (audit fraud, cooler purity)	[member]	Bangladesh retail context
Real fridge footage and blur metric change	[member]	Crete blur effect
Kaggle VLM notebook and prompts (v1, v2)	[member]	Image and video mode
Ground truth and evaluation script	[member]	Draft ground truth
Failure case analysis	[member]	Table II

IV. Weekly Report 4

A. Follow-up from last week

Status of the targets set in Weekly Report 3:

TABLE IV
Keyframe extraction on the 111 supermarket videos (4 fps).

Split	Videos	Duration (s)	Sampled	Kept	Blurry	Duplicate
Train	82	934	3,771	3,420	54	297
Test	29	322	1,298	1,071	115	112



Fig. 1. Real motion blur in two milk-carton videos. Green: kept, red: dropped by the blur filter (blur effect b above 0.5).

1) More real footage — Done. We recorded 111 shelf videos (about 22 minutes) in a Bangladeshi supermarket (Section IV-B1). They show open shelves, not branded fridges.
2) SKU-110K detector — Done. YOLOv8n trained on 2,000 SKU-110K images reaches mAP50 0.913 on 500 held-out images (Section IV-B2).
3) Annotate our own keyframes — Partly done. 222 keyframes were selected and pre-labelled by the detector; correcting the labels is carried over.
4) VLM on product crops — Done (Section IV-B3).
5) Verify the draft ground truth — Not done. Carried over.

B. Work narrative of the current week

1) New footage and dataset statistics: We received 119 video files. SHA-256 hashes showed that 4 were byte-identical copies, and 4 were wide stock clips of whole aisles that we excluded, leaving 111 own recordings (576×1024, median 11 s). They cover personal care, biscuits and snacks, noodles, tea and coffee, juices, soft drinks, cereal, milk, rice and lentils and oil, with TK price tags visible. Neighbouring clips film the same shelf, so we split the videos into train and test sets by blocks of consecutive videos; frames of one shelf never appear in both sets. Table IV gives the extraction statistics at 4 fps.

These videos contain the first real motion blur we have seen. The blur effect ranked frames correctly (sharp frames 0.25–0.35, smeared frames 0.58–0.62, Fig. 1), but frames near the 0.5 threshold can go either way, so we kept 0.5. For annotation, a script picks the sharpest frame in each half of every video: 164 train and 58 test keyframes.

2) Product detector: We trained YOLOv8n [3], [4] with one class, product (no brands), on a random subset of SKU-110K [5]: 2,000 training and 500 validation images. SKU-110K photos are up to about 4,000 pixels wide, so we shrank them once to 1,280 pixels to keep data loading fast. Training ran for 30 epochs at 1,024 pixels on a laptop GTX 1650 (4 GB) and took 163.5 minutes. Table V shows the validation results and two settings that matter for dense shelves; Fig. 2 shows the training curves.

TABLE V
YOLOv8n on 500 SKU-110K validation images.

Setting	mAP50	mAP50-95	Precision	Recall
1,024 px, max. 1,000 boxes (chosen)	0.913	0.564	0.897	0.844
1,024 px, max. 299 boxes	0.889	0.549	0.897	0.833
640 px, max. 1,000 boxes	0.866	0.510	0.875	0.792

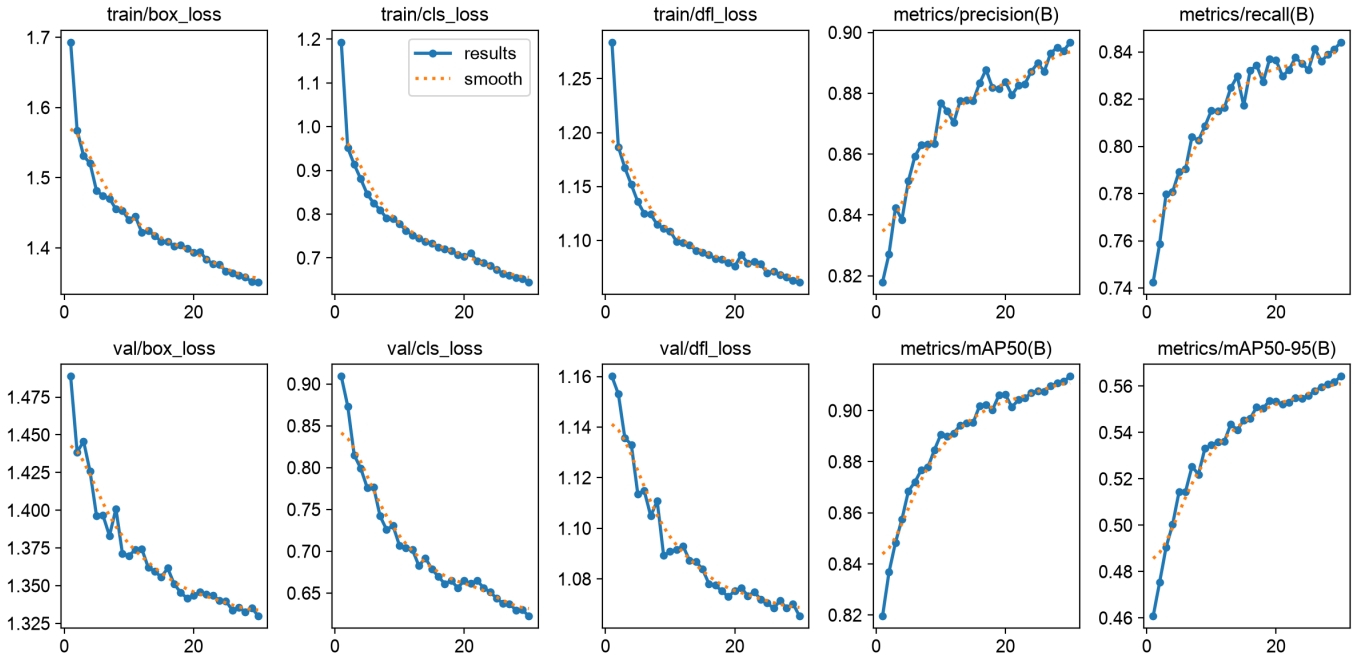


Fig. 2. Training curves. Losses fall on both training and validation data, and mAP is still rising at epoch 30.

TABLE VI
Brand reading on 12 drinks-shelf frames: whole frame vs detector crops (Qwen2-VL-2B).

Method	Usable answers	Brand precision	Brand recall
Whole frame	6 of 12	1.00 (6 frames)	0.30
Detector crops	12 of 12	0.94	0.63

- Detection limit. One validation image holds 718 products and 11 of 500 hold more than 300, so Ultralytics' default limit of 300 boxes per image costs 2.4 mAP50 points. We use 1,000.
- Image size. Products are small, so the default 640 px input costs 4.7 mAP50 points against 1,024 px.
- On our footage. The detector found 151,279 products on our 4,558 keyframes, a median of 29 per frame. It boxes nearly every item on packed shelves seen from a normal distance, but misses large products close to the camera (Fig. 3). Milk-carton close-ups get a median of 6 boxes per frame, against 53 for canned food, because SKU-110K products are small in the image.
- mAP on our own frames needs corrected labels and is carried over. The detector's boxes on the 222 annotation frames (median 30 per frame) are ready to be corrected in Roboflow, which is much faster than drawing every box.

3) Brand reading on product crops: Report 3 concluded that the VLM should be asked about one product at a time. To test this we took the 12 annotation frames of the six drinks-shelf test videos, cut 120 random detector crops, and asked the same Qwen2-VL-2B model (fp16, zero-shot) "which brand is this?" for each crop and "list the brands" for each whole frame. Answers were matched to a catalogue of the 24 expected brands, as an audit for a brand company would; the catalogue was built from our labels, so this is an upper bound. Table VI shows the result and Fig. 4 gives examples.

- Crops doubled the brand recall (0.30 → 0.63) at high precision. Six of the twelve whole-frame answers repeated words until the token limit and gave no usable list, the same loop seen in Report 3.
- Per crop, 48 of 67 readable crops were named correctly (72%; 67% before catalogue matching). Many raw answers are near-misses ("chabac", "canad dry") that the catalogue fixes.



Fig. 3. Detector output on our test videos. Left two: large close-up cartons are missed or split. Right two: packed shelves at normal distance, where nearly every product is found.

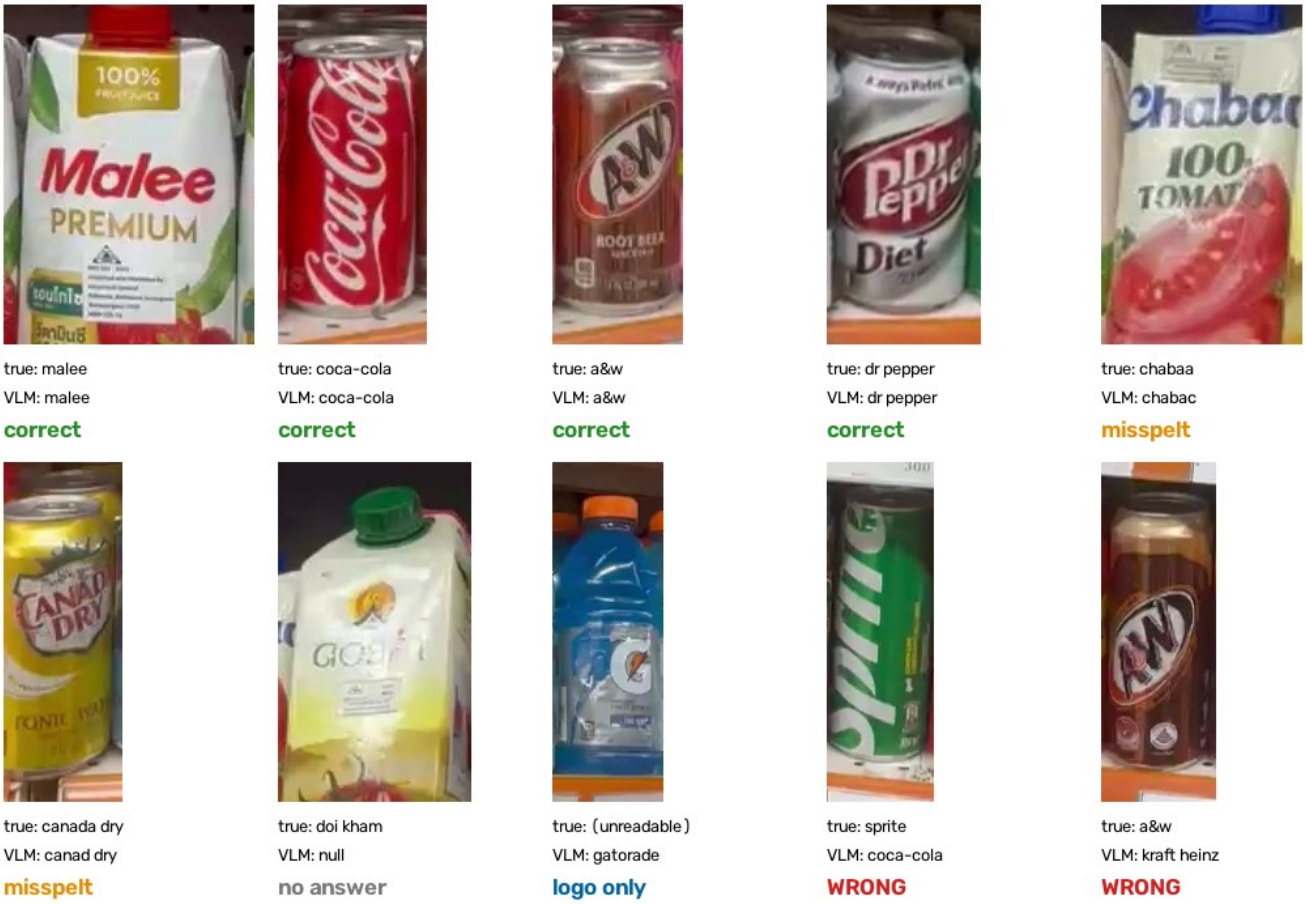


Fig. 4. Examples of single-product crops with the true brand and the VLM answer.

- Wrong brands are rare but biased: 2 of 120 crops were named “Coca-Cola” (for Canada Dry and Sprite), the same pull towards famous brands as in Report 3. A human must still check the evidence crop before a violation is reported.

- No answer was given for 11 readable crops, mostly brands written in Thai or Bengali script.

The crop and frame labels were drafted for this test and still need a second check by the team.

4) Challenges faced:

- The new footage shows open shelves, not branded fridges. It is ideal for product detection, share of shelf and price tags, but cooler purity still needs fridge recordings.
- 4 GB of GPU memory. A batch of 8 images at 1,024 px needs 7.5 GB and a batch of 4 still overflowed on dense images, slowing training 2.7 times; we trained with batches of 2. Each data-loading worker also loaded its own copy of PyTorch, which exhausted Windows memory, so we used one worker.
- A hidden default. Ultralytics silently raises the 300-box limit during validation, which at first made our 300 vs 1,000 comparison meaningless; we re-measured with 299.
- Licence. SKU-110K may be used for academic, non-commercial work only and must not be redistributed, so its photos stay on our machine.

5) Progress made: Sprint 4 is mostly complete: the detector is trained and evaluated on SKU-110K, it runs on all our keyframes, and the crop test supports the hybrid design. The remaining item is mAP on our own frames, which needs the corrected labels.

C. Target for next weekly update

1) Correct the pre-labels on the 222 keyframes in Roboflow, including empty-gap boxes; two members check every test frame.
2) Fine-tune the detector on our 164 train frames and report mAP and counting error on our 58 test frames.
3) Verify the crop brand labels and the Sprint 3 ground truth.
4) Record branded beverage fridges (15–30 s each), including some with competitor products, to test cooler purity.
5) Start Sprint 5: connect extraction, detection and the VLM into one pipeline that writes a JSON audit report with evidence (keyframe, timestamp, box) for every finding.

D. Individual contribution

TABLE VII
Contribution summary.

Task List	Member Assigned	Comments
Supermarket video recording	[member]	115 videos
Video inventory, split and extraction	[member]	4,558 keyframes
SKU-110K detector training and evaluation	[member]	mAP50 0.913
Pre-labels and annotation frame selection	[member]	222 frames
VLM crop test and brand labels	[member]	Recall 0.30 → 0.63

References

[1] P. Wang et al., "Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution," arXiv preprint arXiv:2409.12191, 2024.

[2] F. Crete, T. Dolmiere, P. Ladret, and M. Nicolas, "The blur effect: perception and estimation with a new no-reference perceptual blur metric," in Human Vision and Electronic Imaging XII, ser. Proceedings of SPIE, vol. 6492, 2007, p. 64920I.

[3] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," https://github.com/ultralytics/ultralytics, 2023.

[4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[5] E. Goldman, R. Herzig, A. Eisenschtat, J. Goldberger, and T. Hassner, "Precise detection in densely packed scenes," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019.